

# DEXTER FERNANDES

Computer Vision Engineer   Bristol, UK   +44 7436 007369   [dexterfernandes11@gmail.com](mailto:dexterfernandes11@gmail.com)   [linkedin.com/in/dexterferns](https://linkedin.com/in/dexterferns)  
[github.com/Dexter-Fernandes](https://github.com/Dexter-Fernandes)

## SUMMARY

Computer Vision Engineer with 3+ years developing and deploying object detection, object tracking and perception systems in C++17 and Python. Delivered real-time OpenCV, PyTorch, TensorFlow, DeepStream and OpenVINO pipelines on industrial edge hardware. Verification and validation experience spans repeatable benchmarking, failure-mode analysis and on-site performance assessment. MSc Robotics with hands-on LiDAR SLAM, sensor fusion, ROS2 and embedded deployment.

## RELEVANT EXPERIENCE

### Esbaar - AI Engineer

01/2023 - 08/2023 · MUSCAT, OMAN

- Owned the end-to-end lifecycle for real-time computer vision algorithms on BP drilling rigs, from data curation and PyTorch training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Designed robustness experiments for occlusion, lighting variation and motion artefacts, using TIDE error analysis, ablation studies and repeatable benchmarks to direct dataset and threshold changes.
- Led field verification and validation under motion and environmental stress, turning observed failure modes into model, preprocessing and deployment fixes for live operation.
- Developed and maintained DeepStream 6.x and FFmpeg pipelines with Dockerised services, Kafka, health checks and safe restarts, cutting live video pipeline crashes by 67%.
- Built image, video and sensor-data pipelines with Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout time from 72 hours to under 24 hours.
- Defined a baseline-to-FP16 validation workflow with accuracy, latency, memory and stability acceptance thresholds, and added pytest, unittest and linting to CI for edge inference modules.

### LivNSense Technologies - Lead Software Engineer, AI

05/2020 - 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage object detector and helmet-colour classifier for construction safety, cutting false positives by 33% within latency and power budgets.
- Built asynchronous C++17 perception pipelines with OpenVINO on Intel Movidius VPUs and Core i7 industrial edge systems, raising throughput from 5 to 15 FPS.
- Developed a real-time multi-camera perception system on a moving vehicle and added ByteTrack object tracking to maintain worker and vehicle identities across frames and reduce duplicate alerts.
- Optimised models with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Applied optical-flow ego-motion estimation with PyTorch classifiers to infer vehicle state, and tuned camera and ISP settings for reliable RGB and IR operation.
- Wrote gtest unit tests and documented C++ interfaces, then worked with hardware, software and QA teams to integrate perception modules and support deployment decisions.

## PROJECTS

### DeepStream RTSP Pipeline [github.com/Dexter-Fernandes/deepstream-rtsp-pipeline](https://github.com/Dexter-Fernandes/deepstream-rtsp-pipeline)

06/2026

- Built a DeepStream 9.0 and GStreamer pipeline for YOLO object detection and NvDCF tracking across three concurrent RTSP feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a variable-batch TensorRT FP16 engine, validated accuracy against FP32 and profiled p99 latency to sustain 29.7 FPS per live stream.
- Implemented a custom C++ and CUDA TensorRT decode plugin to move anchor-free post-processing into the inference engine.
- Compared IoU, NvDCF and NvSORT trackers on MOTA, IDF1, identity switches, fragmentation, FPS and VRAM use to support evidence-based tracker selection.

### ROS2 SLAM and Nav2 Simulation

02/2026

- Built a ROS2 Jazzy and Gazebo Harmonic stack to evaluate LiDAR-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations to compare mapping and localisation behaviour.
- Developed C++ and Python ROS2 nodes using tf2, message synchronisation and timestamp alignment, and implemented PCL and Open3D point-cloud filtering, segmentation and registration.
- Added EKF localisation with robot\_localization to fuse wheel odometry and IMU data, and managed reproducible Docker and Git configurations on Linux.

### Modern Edge CV Benchmarking [github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking](https://github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking)

03/2026

- Developed a repeatable performance-assessment harness for PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR, comparing accuracy, latency and model size for edge deployment.
- Exported models to ONNX and evaluated OpenVINO paths, comparing mAP, inference time and model size to support deployment selection.

EDUCATION

University of Bristol - MSc Robotics, Merit

09/2023 - 09/2024 · BRISTOL, UK

Dissertation: Radiological SLAM with LiDAR Odometry

- Built and evaluated a real-time C++17 and ROS2 SLAM system on a Clearpath Husky, integrating Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing through rclcpp and PCL for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer on matched rosbag2 sequences under one evaluation protocol, with Ceres, g2o and GTSAM solver backends and tuned ICP parameters across environments.
- Calibrated LiDAR extrinsics to the robot base, validated sensor placement for coverage and resolved transform timing with tf2 across a registered point-cloud pipeline.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier under JetPack within compute and memory limits, and tested Nav2 survey routes in simulation before running them on the Husky.

VNIT - BTech Electrical and Electronics Engineering

07/2016 - 05/2020 · NAGPUR, INDIA

- Built foundations in signals, systems, embedded hardware and control supporting later work in real-time perception, sensor systems and robotics software.

SKILLS

|                    |                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages          | Python, C++17, CUDA C++, Bash, SQL                                                                                                                                                                          |
| Computer Vision    | Object detection, object tracking, image classification, semantic and instance segmentation, optical flow, data augmentation, OpenCV, PyTorch, TensorFlow, Keras, scikit-image, YOLO, RT-DETR, EfficientDet |
| Robotics & Sensors | ROS2 (Robot Operating System 2), rclcpp, tf2, rosbag2, LiDAR SLAM, sensor fusion, LiDAR extrinsics calibration, PCL, Open3D, RTAB-Map, LIO-SAM, Cartographer, Nav2, EKF, ICP, pose graph optimisation       |
| Optimisation       | TensorRT, OpenVINO, ONNX, FP16/INT8, quantisation-aware training, structured pruning, layer fusion, Intel VTune, latency, throughput, memory and VRAM profiling                                             |
| Video Systems      | NVIDIA DeepStream, GStreamer, FFmpeg, RTSP, NvDCF, NvSORT, ByteTrack, multi-stream video analytics, multi-camera synchronisation, metadata extraction, RGB and IR camera tuning                             |
| Verification       | Performance assessment, automated testing, benchmarking, acceptance thresholds, robustness testing, failure-mode analysis, ablation studies, hypothesis testing, TIDE, field validation                     |
| Embedded           | NVIDIA Jetson AGX Xavier, JetPack, Intel Movidius Myriad VPU, Intel Core i7 industrial edge systems, multiprocessing, compute and memory constraints                                                        |
| Experimentation    | MLflow, DVC, Optuna, Weights & Biases, CVAT, synthetic data generation, dataset versioning, Bayesian optimisation, drift tracking                                                                           |
| Software Delivery  | Git, CI/CD, gtest, pytest, unittest, CMake, Docker, Linux, Kafka, REST APIs, structured logging, health checks, production monitoring                                                                       |

ADDITIONAL EXPERIENCE

GrowthStage - Technical Assessor, Computer Vision and Software Engineering

05/2026 - PRESENT · REMOTE

Assess candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019)

Best Speaker Award, ABC Toastmasters Club (01/2020)